Pradeep Gandu



Cloud Data Engineer | Automotive Software | AI / Machine Learning

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LinkedIn: linkedin.com/in/pradeep-gandu | GitHub: github.com/gandupradeep2026 | Portfolio: gandupradeep2026.github.io

German: C1 (UNIcert III / DSH) | English: Fluent (C1) | Status: § 16b AufenthG (Full-Time from 01/2027)

EXECUTIVE PROFILE

Cloud Data and Automotive Software Engineer with practical experience designing event-driven big data streaming pipelines (GCP BigQuery, Pub/Sub, Apache Beam, Dataflow) and deploying generative AI / RAG architectures (FastAPI, MCP, Ollama at DiPP GmbH). Master's candidate at Chemnitz University of Technology with professional C1 German proficiency, nationwide mobility, and immediate availability for working student/part-time roles (full-time from January 2027).

CORE COMPETENCIES

- Cloud Data Engineering (GCP / AWS) • Apache Beam / PySpark / Spark SQL • Vehicle Telemetry & CAN-Bus (ISO 14229)
- CARLA AD Simulator & Scenario Runner • Python / FastAPI / REST APIs • Generative AI / RAG / MCP / Ollama
- BigQuery / Cloud Storage / Cloud Composer • CI/CD GitHub Actions / Docker / Linux

PROFESSIONAL EXPERIENCE

Working Student — AI Software Solutions — DiPP GmbH

Chemnitz, Germany — **01/2025 – 07/2025**

Technologies / Stack: Generative AI, FastAPI, RAG, Model Context Protocol (MCP), Ollama, IONOS Cloud, Docker, Git

- Engineered and integrated generative AI and LLM services into enterprise CRM systems via Python and FastAPI
- Built modular RAG pipelines covering document ingestion, text chunking, vector embeddings, and semantic retrieval
- Standardized application context and external tool interfaces via Model Context Protocol (MCP) for secure local inference
- Conducted integration testing, automated error logging, and systematic response evaluations for client deployments

Associate Language Specialist (Content Engineering & NLP) — GlobalLogic Technologies

Hyderabad, India — **10/2020 – 12/2021**

Technologies / Stack: German Retail Data, NLP / Text Processing, Machine Learning, scikit-learn, PyTorch, Python

- Preprocessed, curated, and validated German retail e-commerce datasets for automated machine learning pipelines
- Trained, tuned, and evaluated supervised classification models using Python, scikit-learn, and PyTorch
- Conducted systematic data quality audits to guarantee high annotation consistency according to client guidelines

Senior Process Executive (ML Quality Assurance Meta / Facebook) — Cognizant

Hyderabad, India — **09/2017 – 05/2019**

Technologies / Stack: Facebook Marketplace, Human-in-the-Loop ML, Data Quality, SQL, Python, Root Cause Analysis

- Performed Human-in-the-Loop ML evaluation and metadata validation for Facebook Marketplace categorization
- Improved classification accuracy from approximately 64 % to 76 % through structured feedback loops
- Executed automated data quality audits and root cause investigations for model anomalies using SQL and Python

Practitioner — CRM Operations (Quality Assurance for Google) — Concentrix Daksh Services

Hyderabad, India — **03/2016 – 05/2017**

Technologies / Stack: Google Ads, Workflow Automation, Process Validation, Error Analysis, SQL

- Conducted process validation, error analysis, and data verification for automated Google Ads workflows
- Raised operational process success rate from approximately 72 % to 79 % through workflow optimization

EDUCATION

M.Sc. in Automotive Software Engineering — **Chemnitz University of Technology**
Germany — **10/2022 – expected 12/2026**

- 90 / 120 ECTS completed; Master's thesis in finalization stage: Generation and Annotation of Scenario-Centric Datasets for Environment Prediction in Autonomous Driving (Python, CARLA Simulator 0.9.15, OpenDRIVE, Parquet)
- **Key Areas:** Distributed Systems, Software Architectures, Advanced Driver Assistance Systems, Simulation & HIL

M.Tech. in Embedded Systems — **MLR Institute of Technology**
Hyderabad, India — **09/2017 – 09/2019**

Graduated with Distinction: First Class with Distinction (CGPA: 7.33 / 90 Credits) | **Key Areas:** RTOS, Microcontrollers, System Design

B.Tech. in Electrical & Electronics Engineering — **Jawaharlal Nehru Technological University**
Hyderabad, India — **08/2009 – 11/2013**

Degree: First Class (Aggregate: 62.67 %) | **Key Areas:** Control Systems, Electronics, Signal Processing

SELECTED ENGINEERING PROJECTS

Automotive Telemetry Streaming Platform (GCP) — **2026**
Stack: GCP Pub/Sub, Apache Beam, Dataflow, BigQuery, Airflow, Python — **GitHub**

- Built scalable event-streaming pipeline for real-time CARLA vehicle telemetry with stateful deduplication via Apache Beam
- Configured partitioned, cost-optimized BigQuery data warehousing and orchestrated automated DAGs via Cloud Composer

AutoSense AI: Predictive Vehicle Diagnostics & Agent — **2026**
Stack: FastAPI, IsolationForest, RAG, Ollama LLM, MCP, React / Vite — **GitHub**

- Designed anomaly detection on 39,000+ real OBD-II records with a 5-out-of-7 stabilization filter reducing false alarms by 93.4 %
- Implemented interactive diagnostic agent querying technical workshop repair manuals via RAG and Model Context Protocol

TU Chemnitz Badminton Community Portal — **2026**
Stack: React / Vite, Node.js / Express, SQLite, Tailwind, Cloudflare, Oracle VM — **Live App**

- Architected decoupled full-stack platform with React/Vite SPA on GitHub Pages and Express/SQLite backend via Cloudflare Tunnel
- Built tournament registration, partner matching, trainer directory, and hardened admin security with cryptographic OTPs

TECHNICAL SKILLS & CERTIFICATIONS

Cloud & Big Data: Google Cloud Platform (BigQuery, Pub/Sub, Dataflow, Dataproc, GCS, IAM), AWS (EC2, S3), Apache Beam, PySpark, Spark SQL, Apache Airflow, Parquet
Software & AI: Python (FastAPI, Scikit-learn, PyTorch), SQL (Advanced), RAG Architectures, Ollama, MCP, REST APIs, Docker, Docker Compose, Git / GitHub Actions CI/CD
Automotive & IT: CARLA AD Simulator, OpenDRIVE, 3D Trajectories, CAN-Bus, ISO 14229 (UDS), Linux (Ubuntu/Debian), Windows Server, Active Directory
Certifications: Google Cloud Certified Data Engineer • UNIcert III C1 German (TU Chemnitz) • Cisco Python • HackerRank SQL

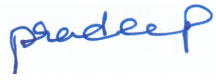
LANGUAGES

German: C1 (UNIcert III / DSH, Professional Working Proficiency) | **English:** C1 (Fluent) | **Telugu:** Native | **Hindi:** Fluent

VOLUNTEERING & INTERESTS

Badminton: Active player in Chemnitz University of Technology university sports; founder and developer of the community web portal
Intercultural Exchange: Community support and peer networking for incoming international Master's students in Chemnitz

Chemnitz, 14 September 2026



Location, Date

Pradeep Gandu (Signature)